Corridor Wealth Technical Document

Proof-carrying wealth advice with inspectable decisions and controlled execution

Team name	{{TEAM_NAME}}
Competition track	International AI Track — Topic B: Wealth Advisory Agent
Submission date	2026-10-20

01 Project Overview

Wealth advice that remains inspectable at execution

Corridor Wealth is a synthetic advisory and purchase workflow for International AI Track Topic B. A client clarifies goals, reviews a complete declared product comparison, receives a policy-checked allocation and separately approves each purchase. The model proposes; an independent checking path decides whether the proposal may execute.

The motivating failure is a persuasive recommendation that cannot justify its numbers, suitability or authority. A model may omit a cheaper eligible product, repeat an issuer's promotional claim, or treat a malicious document as an instruction to buy. Corridor requires financial facts and permissions to survive deterministic checks even when model output is hostile.

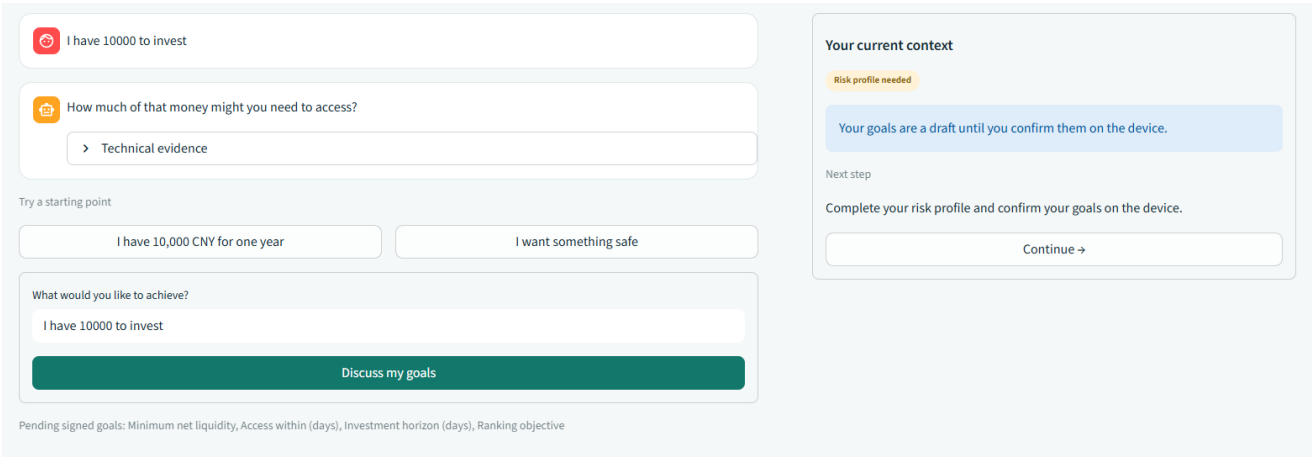
The Best-Interest Allocation Certificate, or BIAC, carries the signed context, eligible alternatives, selected amounts, costs and constraints needed to recompute a proposal. The client can inspect what was excluded and why. The gateway then binds approval to the current canonical facts and checks again before execution.

Claim boundary. The allocation is a **policy-checked heuristic**, and is **sponsor-independent within the declared catalogue and policy**. These are technical properties of a bounded synthetic workflow, not a legal best-interest certification. All institutions, products, accounts and people use **synthetic data**. Authentication uses **simulated MFA**. This prototype is **not investment advice**.

What the judge can inspect

The following pages connect the client workflow to its trust boundary, certificate computation, approval lifecycle and reported evaluation. The companion security assessment states threat assumptions and residual risks. The execution evidence document distinguishes retained F5 measurements from retained I1 scripted UI evidence captured before W1 and records current offline replay verification.

Topic B is an end to end client workflow



Current Advisor clarification screen. Scripted UI evidence using synthetic inputs; no live-model result is implied.

Required capability	Client experience	Inspectable mechanism
Product comparison	Review eligible and excluded products, fees and liquidity	Complete declared catalogue with deterministic rankings
Client risk profiling	Review and sign questionnaire answers on the device	Freshness, consumption, scoring and effective suitability checks
Advisory dialogue	Clarify missing goals; ask why; correct or stop	Unsigned draft slots, sourced facts and explicit state transitions
Explainable allocation	Review weights, costs and differences before selection	Signed policy and recomputable BIAC

The prototype allocates new money from a CNY funding account for a single synthetic client. Existing holdings do not enter the allocation-weight denominator. Historical holdings still remain visible after a partial purchase. There is no automatic rebalancing, real financial account connection or market forecast.

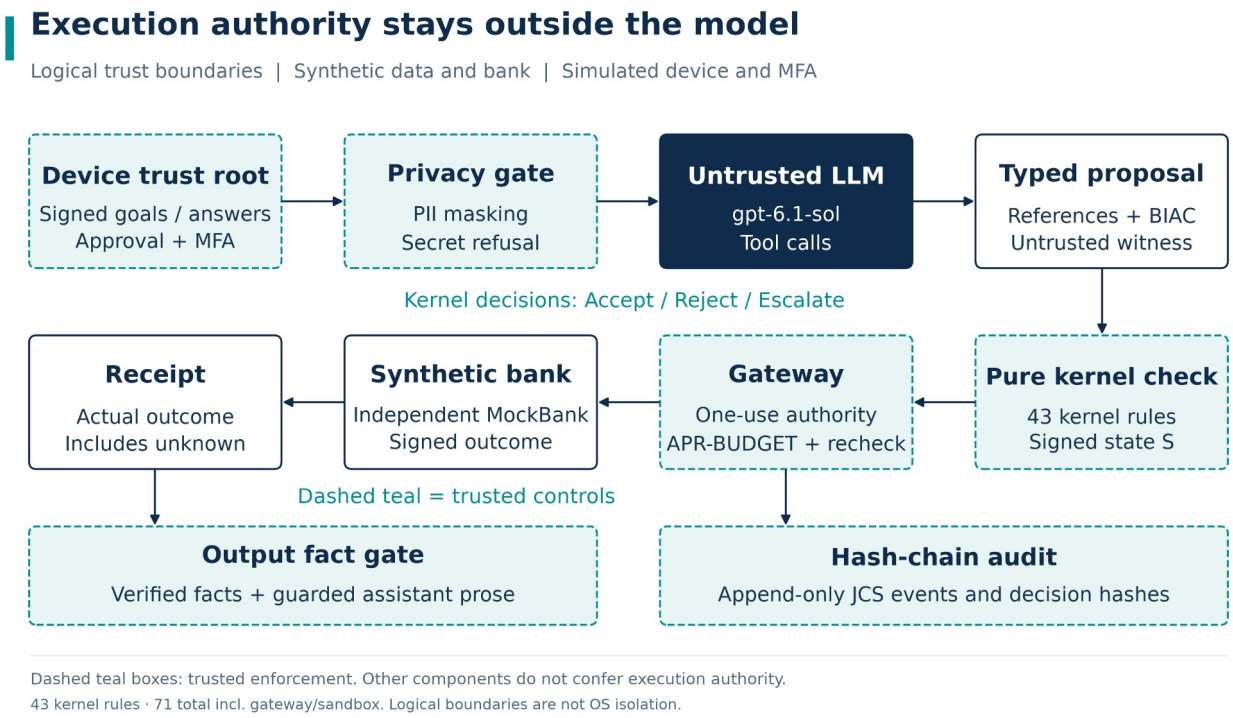
The official Topic B wording asks for a professional-style workflow. Corridor implements the interaction and evidence chain within a synthetic domain; it does not equate a short questionnaire or a certificate name with professional suitability.

Source: official topic requirements, accepted wealth design decisions, and current workflow source.

02 System Architecture

The trust boundary separates proposals from authority

model proposes · kernel decides · device authorises · gateway executes



Architecture and trust boundary. Rule counts are source inventory, not experimental outcomes. Model-side content cannot supply signatures or executable money literals.

The application trusted computing base comprises schemas, crypto, kernel, stores and gateway. The pure kernel reads trusted state and checks proposals without a model client or network dependency. The gateway resolves permitted references, renders the approval card, validates device evidence, serializes dispatch and writes the result.

The untrusted dialogue and planner may propose tool calls and typed references. The configured primary model is gpt-6.1-sol. Its role is to clarify intent and choose available operations; it cannot sign an input, issue an approval, fabricate a receipt or bypass the gateway. Input minimization precedes model calls. A fact gate and output guard constrain what the application presents as verified financial information.

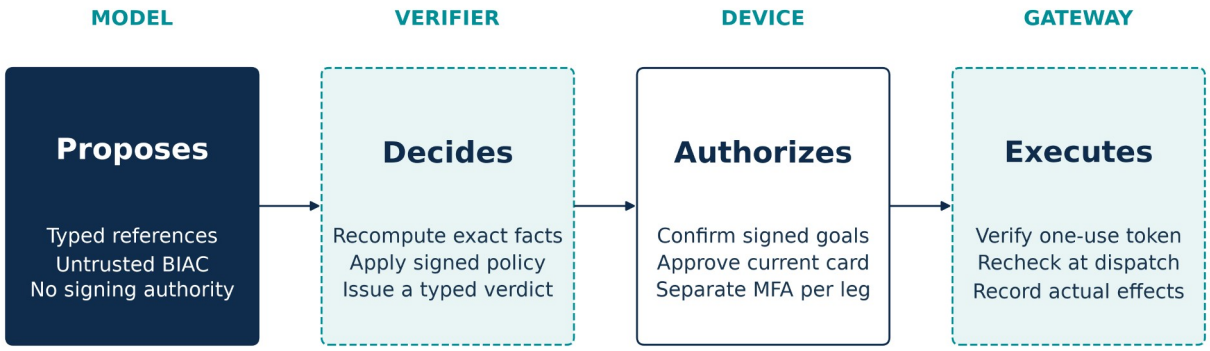
The synthetic bank independently executes submitted operations and signs its receipts. The reference oracle does not import the production kernel or stores. This independence helps reveal shared-code errors, but a mistaken shared specification remains possible. The in-process Python boundary is logical isolation; it does not protect against hostile code with host memory or signing-key access.

Source: rule and dependency inventory, import contracts, and AST boundary tests.

Authority has distinct owners

The model proposes; a verifier decides

Separate intelligence, decision, consent and execution



A plausible answer is never a permission to move funds.

Synthetic data • Simulated MFA • Not investment advice
Implementation: kernel/check.py; gateway/service.py; sim/device.py

Authority split. The model proposes, the kernel decides, the device authorises and the gateway executes. These roles remain distinct even when hosted in one process.

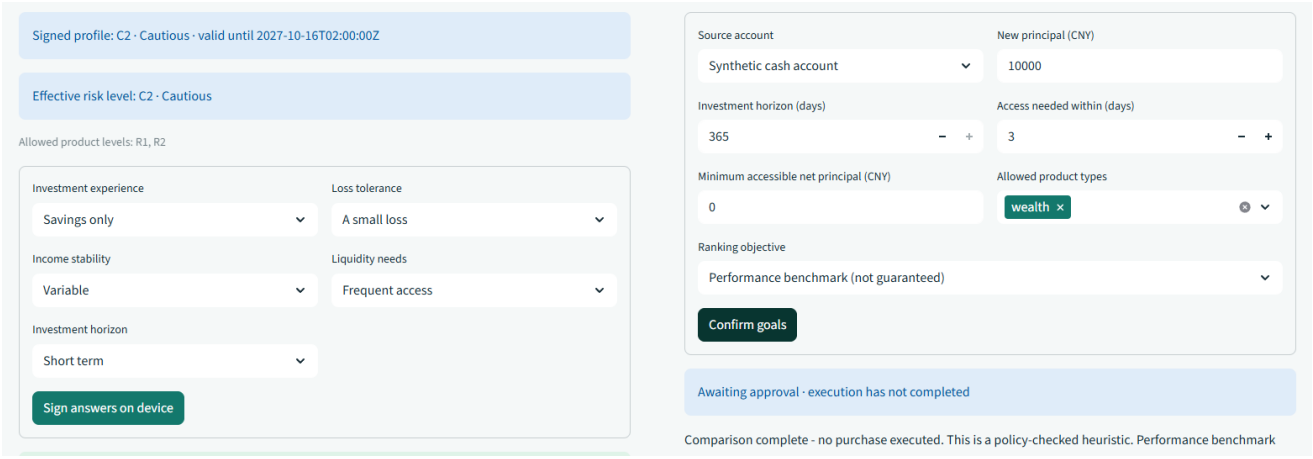
Component	Responsibility	It does not establish
Device and signed state	Explicit goals, answers, policy and operation scope	Real identity, civil capacity or production key management
Planner and prover	Typed proposal and candidate certificate	Correct arithmetic, client consent or execution permission
Pure kernel	Recompute provenance, costs, suitability and constraints	Truth of a signed catalogue or global allocation optimality
Gateway and bank	Current-card approval, dispatch checks and signed effects	Atomic completion of an entire allocation

Money remains exact as rational values in memory and reduced rational strings on the wire. Executable amounts require whole cents. JSON floating-point values are rejected; display rounding does not alter signed values. Business time comes from the simulated clock, while deterministic identities keep repeated requests reproducible.

Device inputs, policies, catalogues, approvals and receipts use distinct signing contexts. Demonstration keys derive from a deterministic labelled seed. They are suitable for repeatable synthetic tests and unsuitable for production key management. See cryptography implementation and demo key derivation.

03 Agent Design

A conversation becomes authority only after device confirmation



Device-signed goals in the retained I1 UI capture. Scripted evidence; goals and questionnaire answers remain separate explicit client actions.

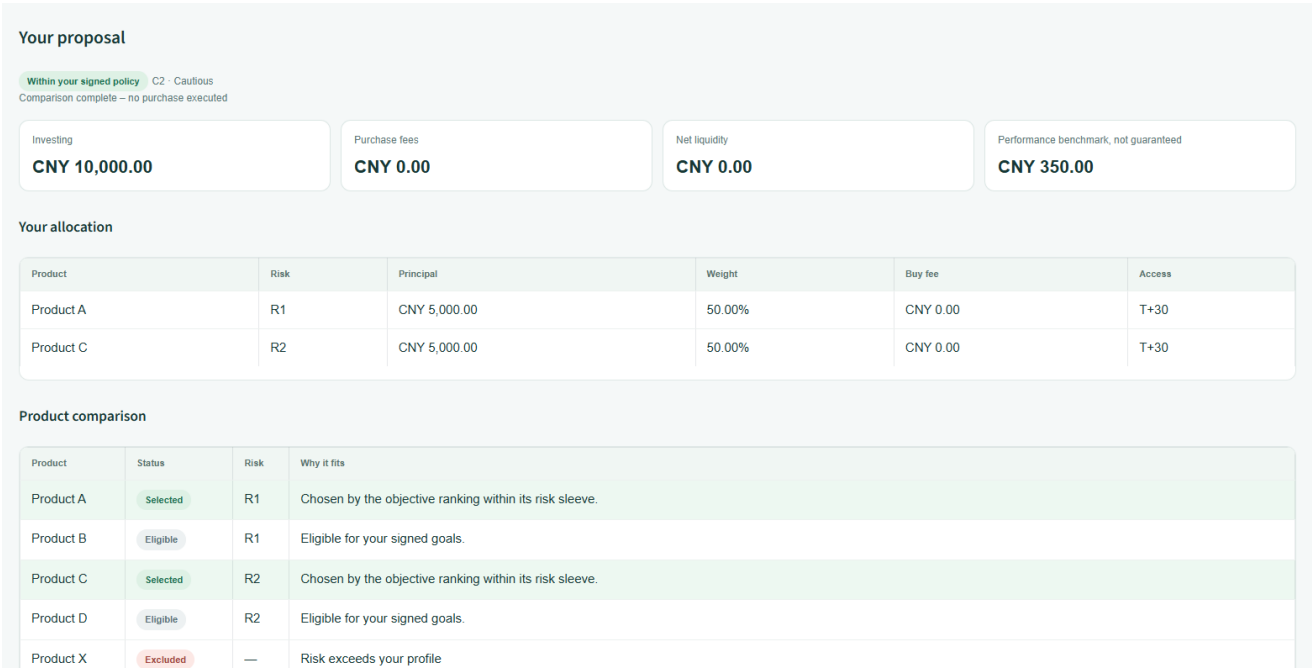
The dialogue accumulates explicitly sourced, unsigned slots for amount, horizon, liquidity need and objective. Missing or ambiguous values cause clarification. A phrase such as "something safe" cannot set a risk grade or become an executable purchase amount. The device reviews the completed goal sheet and signs the structured request.

Questionnaire answers remain outside model authority. Raw answers and scoring evidence are preserved; effective suitability can be tightened by signed goals and trusted session signals. A higher raw grade cannot erase a tighter effective cap. Requests to coach answers toward a target rating are refused and audited.

Read-only explanation turns preserve an existing execution or reconciliation result. Stop withdraws authority before optional model work, so a model failure or strict-cache miss cannot prevent cancellation. Corrections supersede old inputs and terminate relevant pending plans; the new goal requires fresh device confirmation.

All model calls pass through the guarded LLM entry point. The API profile and bounded output settings are part of replay identity. A strict miss fails; it cannot silently substitute a scripted response into a live-model result.

The BIAC exposes the complete declared comparison



Your proposal

Within your signed policy · C2 · Cautious

Comparison complete – no purchase executed

Investing	Purchase fees	Net liquidity	Performance benchmark, not guaranteed
CNY 10,000.00	CNY 0.00	CNY 0.00	CNY 350.00

Your allocation

Product	Risk	Principal	Weight	Buy fee	Access
Product A	R1	CNY 5,000.00	50.00%	CNY 0.00	T+30
Product C	R2	CNY 5,000.00	50.00%	CNY 0.00	T+30

Product comparison

Product	Status	Risk	Why it fits
Product A	Selected	R1	Chosen by the objective ranking within its risk sleeve.
Product B	Eligible	R1	Eligible for your signed goals.
Product C	Selected	R2	Chosen by the objective ranking within its risk sleeve.
Product D	Eligible	R2	Eligible for your signed goals.
Product X	Excluded	—	Risk exceeds your profile

Complete proposal and BIAC in the retained I1 UI capture. Scripted evidence using synthetic data and simulated authentication; factual fields and source links remain inspectable.

The certificate exposes the selected products, risk bands, costs, constraints and source bindings together. Its checks are technical statements under the declared synthetic catalogue and policy, not a real-world suitability certification.

Recomputing the certificate keeps its claims bounded

The kernel checks the current signed policy, catalogue and risk record before constructing the eligible set. Eligibility asks whether a product can accept an allowed candidate amount. This avoids excluding a useful partial holding merely because that product cannot accept the entire request principal. Every exclusion retains its applicable reasons.

Eligible products are ranked at a common reference amount and horizon. That ranking is explicitly comparison-only. Actual purchase amounts are priced separately, and fees are rounded upward per operation before aggregation. All displayed return estimates carry **performance benchmark (not guaranteed)**.

BIAC makes recomputable	BIAC does not prove
Complete eligible set within the signed catalogue	Market-wide availability or an unbiased catalogue
Exact amounts, costs and net-liquidity checks	Future realized returns or liquidity under market stress
Signed risk bands, concentration and explicit deviations	Whole-household diversification or legal suitability
Binding from goals to the selected parent plan	Global optimality or atomic completion of every purchase

Source: allocation kernel and accepted design decisions.

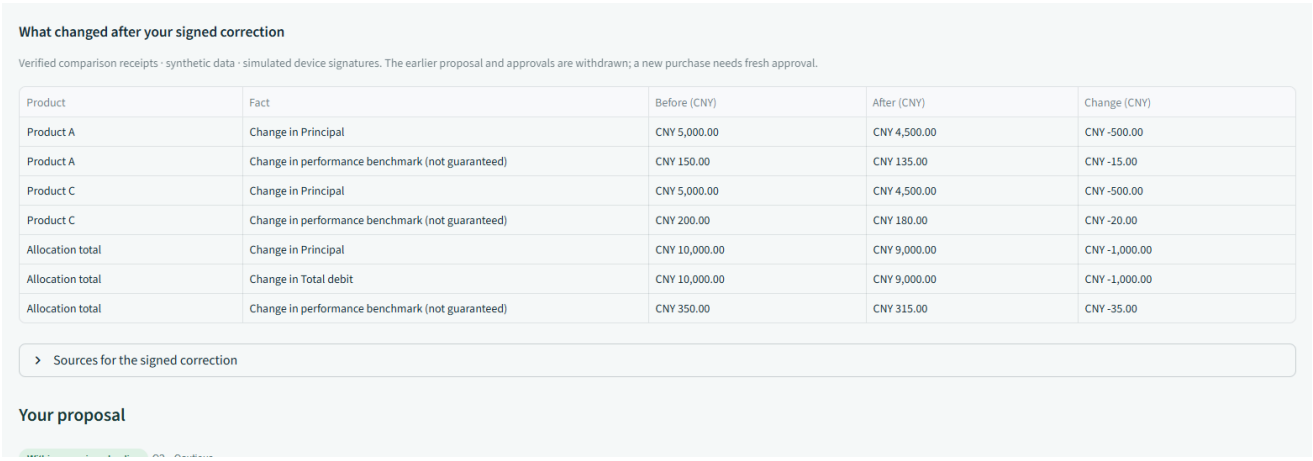
The allocation algorithm fails conservatively

The selector is a bounded greedy construction. It allocates within signed risk-sleeve budgets, first addressing the net-liquidity requirement and then filling remaining capacity. It finally recomputes principal conservation, amount resolution, product limits, concentration, individual and group bands, fees, available funds and liquidity. The certificate cannot replace these recomputations with a claimed pass flag.

Accessible value is net principal after the synthetic contract's holding and redemption fees by the signed deadline. Projected return is excluded from this liquidity amount. Cash left outside the requested investment does not count toward the invested-principal requirement.

Failure is not an infeasibility proof. A feasible allocation can be missed by this heuristic. The correct outcome is "No allocation was constructed under this policy and these inputs." Construction failure remains a completion cost and does not authorize a weaker policy or an unverified alternative.

Client corrections remain explicit and visible



What changed after your signed correction

Verified comparison receipts · synthetic data · simulated device signatures. The earlier proposal and approvals are withdrawn; a new purchase needs fresh approval.

Product	Fact	Before (CNY)	After (CNY)	Change (CNY)
Product A	Change in Principal	CNY 5,000.00	CNY 4,500.00	CNY -500.00
Product A	Change in performance benchmark (not guaranteed)	CNY 150.00	CNY 135.00	CNY -15.00
Product C	Change in Principal	CNY 5,000.00	CNY 4,500.00	CNY -500.00
Product C	Change in performance benchmark (not guaranteed)	CNY 200.00	CNY 180.00	CNY -20.00
Allocation total	Change in Principal	CNY 10,000.00	CNY 9,000.00	CNY -1,000.00
Allocation total	Change in Total debit	CNY 10,000.00	CNY 9,000.00	CNY -1,000.00
Allocation total	Change in performance benchmark (not guaranteed)	CNY 350.00	CNY 315.00	CNY -35.00

› Sources for the signed correction

Your proposal

Goal correction and comparison differences in the retained I1 UI capture. Scripted evidence; revised goals require fresh device confirmation and invalidate earlier approval.

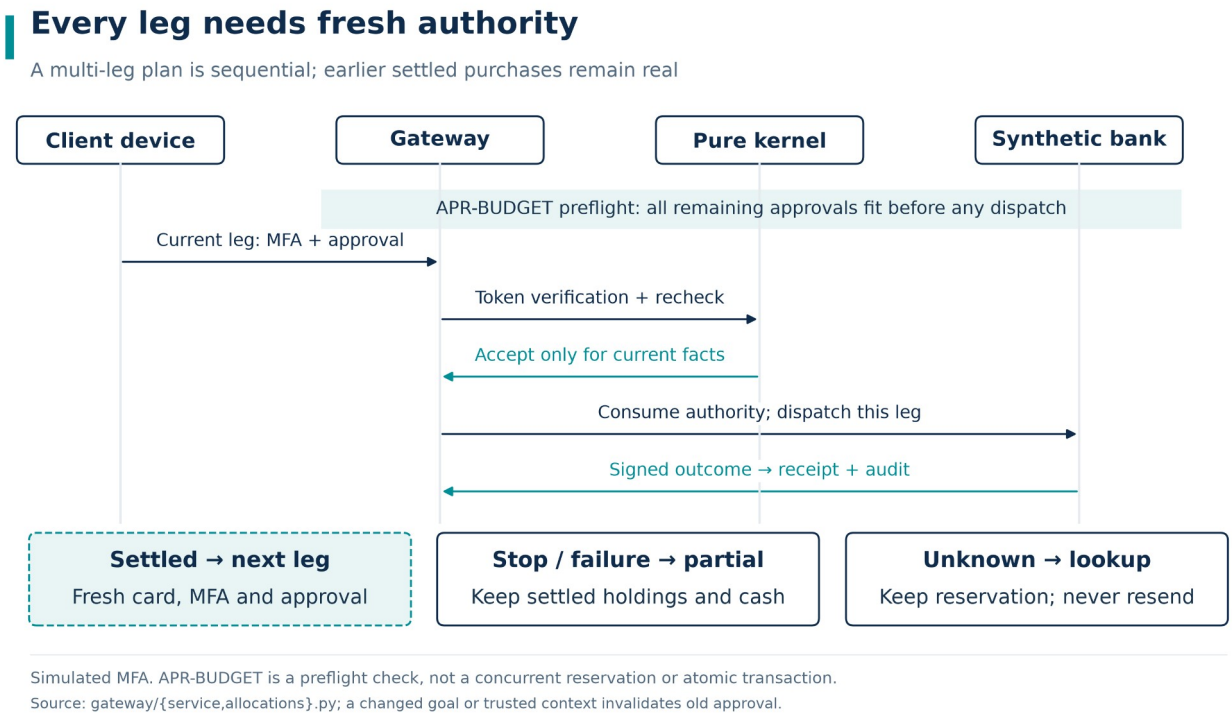
Explicit weights or selected products bind to a reconstructible default in the same economic context. The card retains the deviation and exact changes in amount, weight, fees, benchmark and liquidity. An unsigned what-if preview grants no authority. Selecting a preview must pass ordinary device signing and kernel validation again.

The comparison view connects each displayed change to authenticated before-and-after facts. A lower principal changes product amounts and derived benchmarks, while the old proposal and its approvals are withdrawn. The client can review the effect before deciding whether to sign and approve the revised plan. Chat does not make that decision for them.

A changed goal cannot erase a settled purchase. If execution has started, the workflow preserves the actual parent result and its holdings. Stop blocks later legs before optional model work. A new request requires a new device-confirmed intent; it cannot reuse a terminal plan or its consumed authority. This keeps correction, cancellation and reconciliation distinct and visible.

04 Security Design

Every purchase is checked at the moment it can move money



Allocation approval sequence. APR-BUDGET is checked before the first leg; each leg retains its own current card, simulated MFA and one-use approval.

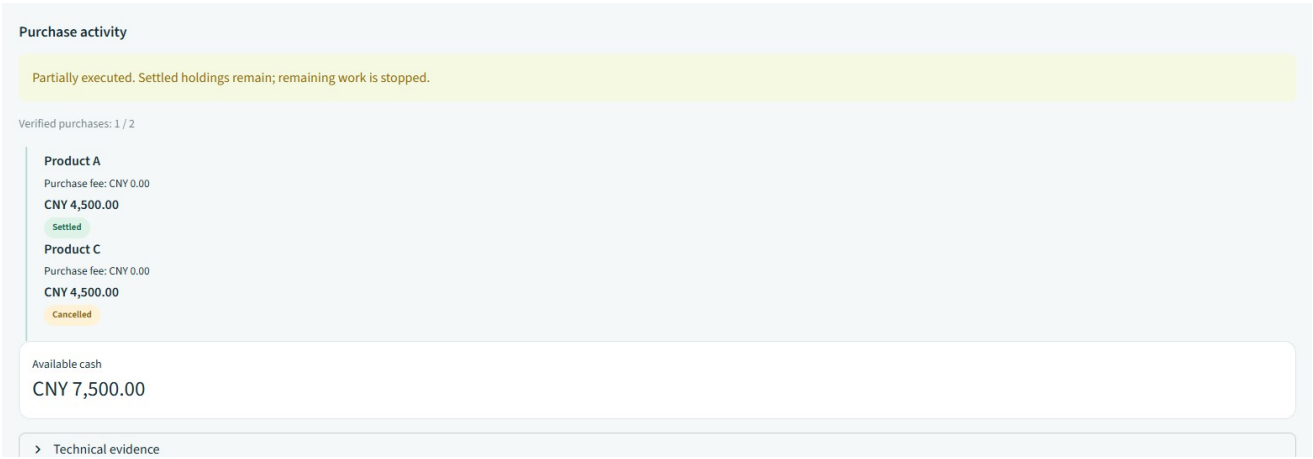
The device selects the complete intended positive allocation and derives BIAC-bound child intents and separate authorizations. Each purchase receives a current canonical card, its own scoped simulated MFA grant and device approval. The gateway verifies the approval token and reruns the pure check in dispatch phase under its lock.

`AUTH-CONSUMED` prevents reused authority after reserved, unknown or settled effects. Approval nonce consumption is separate from a business reservation. A failed attempt may release reserved funds without making the same approval reusable. Material card drift still rejects.

`APR-BUDGET` checks whether approval capacity covers the remaining plan before the first dispatch. An unchanged propose/MFA lifecycle shares its count; distinct or drifted requests still count. This is a preflight check, not a reservation of future capacity: concurrent activity can still stop a plan after an earlier leg succeeds.

Source: approval budget regressions, gateway orchestration, and rule inventory.

Truthful recovery preserves the effect that actually happened



Partial completion after Stop in the retained I1 UI capture. Scripted evidence; the settled holding remains visible while the later purchase is cancelled.

Purchases are not atomic as a group. Stop, failure or handoff blocks later legs while retaining settled holdings. An unknown result retains its reservation and requires a signed lookup; the system does not resend an order to manufacture certainty. A correction cannot erase an earlier effect or silently restart a terminated plan.

Later legs reconstruct the original budget using only this parent's verified settled debit. Unrelated spending, changed policy or catalogue, expired risk or superseded inputs can invalidate the remaining operation. Dependencies are scoped to the relevant principal and parent, preventing unrelated plans from invalidating a legitimate card.

Private or secret-bearing utterances are rejected before model calls. Model-visible state uses minimized role-filtered handles. Verified receipt references preserve their original source paths; raw model numbers and guarantees cannot become verified narration. Finite privacy and output guards remain bounded checks, not proof against all encodings or meanings.

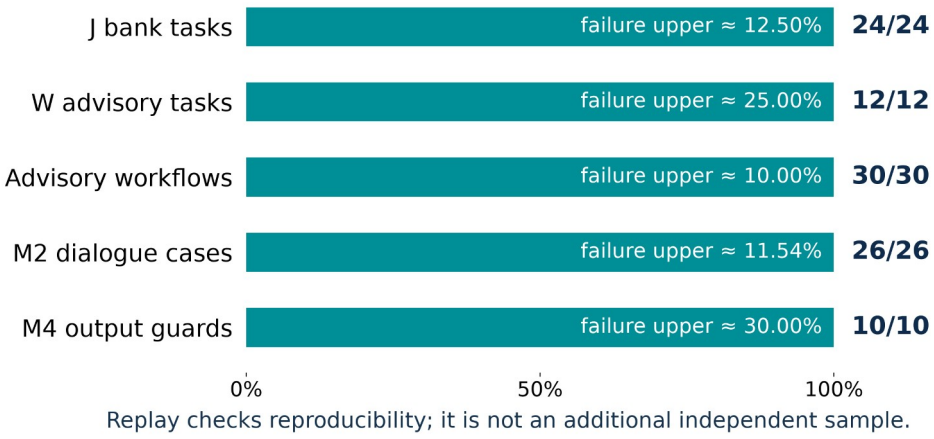
Audit records append canonical bytes, retaining decisions, rule identifiers and commitments. A write failure blocks protected execution. Chain verification assumes the custodian preserves a trusted head or checkpoint; replacing every local copy is outside that guarantee.

05 Deployment and Testing

Completion and replay answer different questions

Recorded completion reproduces offline

gpt-6.1-sol • F5 live-recorded outcomes • historical replay on code before I1



W live results cover disclosed advisory workflows; formal mutation probes remain scripted.
Original F5 evidence retained. Descriptive 95% failure bounds. Source: FINAL-RESULTS.

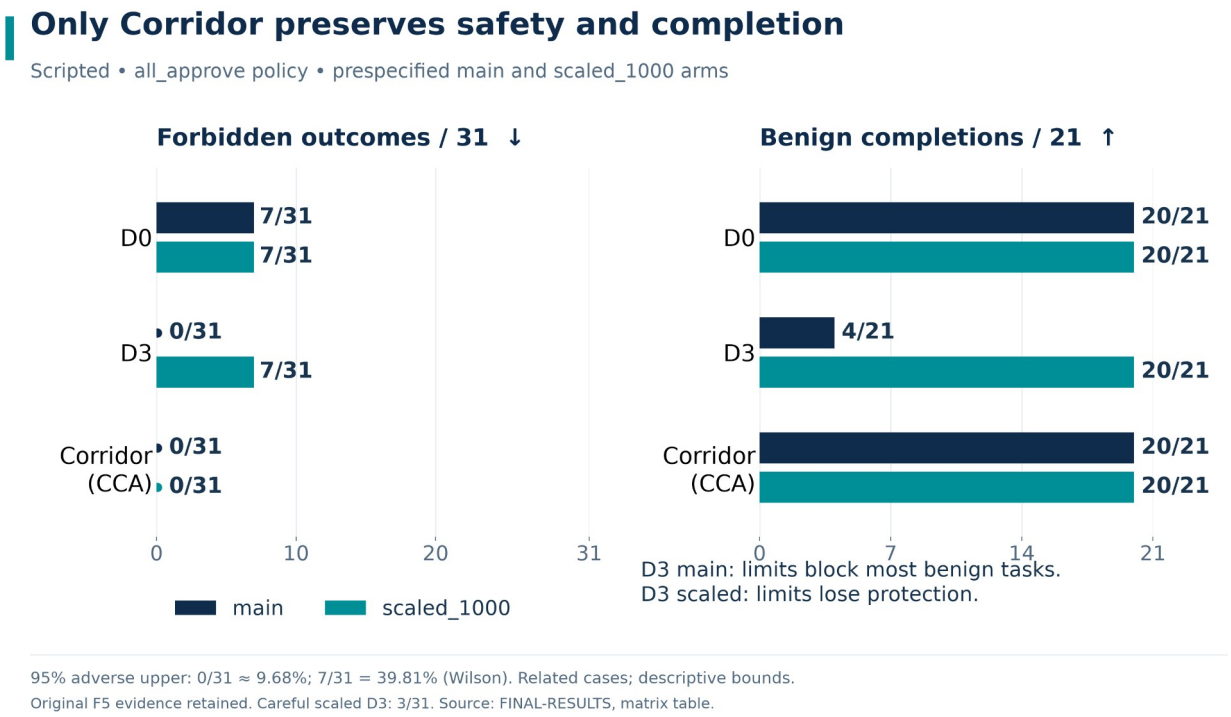
Reported F5 task completion. Live-recorded results and transport-disabled strict-replay agreement are retained-report evidence; replay is not an independent second sample.

The retained FINAL-RESULTS report records gpt-6.1-sol live-recorded completion of J tasks at **24/24**, W tasks at **12/12**, and advisory workflows at **30/30**. It reports M2 dialogue checks at **26/26** and M4 output checks at **10/10**, with corresponding strict-replay agreement. Formal W constraint probes remain scripted and are not added to the live workflow denominator.

For these zero-failure rows, the report's descriptive approximate upper bounds at 95% are respectively **12.5000%**, **25.0000%**, **10.0000%**, **11.5385%** and **30.0000%**. Related cases are not an independent population sample. M4 uses the same output guard for observed checks and does not establish an independent unsafe-output oracle.

The original report and complete evidence/final-f5 recordings are retained. These figures report the F5 observations. Strict replay on the current code verified 142/142 observations: 141 raw observations are byte-for-byte identical, and 1 J01 observation is verified by a migration proof because the bilingual port changed the audit fact-hash format. Network transport was disabled and no API calls occurred. P2's historical 142/142 byte-for-byte equality used code before I1. Replay reproduces recorded failures as well as successes; it is not an independent model sample. The execution evidence document provides the command and result summary.

Limits alone do not preserve useful completion across scale



Scripted baseline matrix from the retained F5 report. Prohibited outcomes use all execution projections; benign completion has its own denominator.

Under the **scripted all_approve policy**, Corridor CCA reports **0/31** prohibited outcomes and **20/21** benign completions in both main and scaled_1000. D3 reports **0/31** and **4/21** in main, but **7/31** and **20/21** in scaled_1000. D0 reports **7/31** and **20/21** in both. The comparison isolates a mechanism trade-off in these prescribed projections, not general performance against arbitrary financial agents.

The retained report gives a descriptive approximate 95% upper bound of **9.6774%** for **0/31** prohibited outcomes. The **7/31** Wilson upper bound is **39.8124%**. Benign failure upper bounds are **22.6694%** for **20/21** completion and **92.3324%** for **4/21**. The partial-stop projection remains incomplete by design.

The careful-user arm is disclosed separately: scaled_1000 D3 has **3/31** prohibited outcomes, rather than the all_approve arm's **7/31**. Its benign completion remains **20/21**. Accepted forged proposals count as prohibited outcomes even before money moves. Missing-MFA protocol gaps are reported separately from business harm. No favorable arm is silently substituted for another.

External validation: AgentDojo banking

Security must not depend on the model resisting injection: with the planner fully hijacked in model-free GroundTruth replay on AgentDojo banking v1.2.2 (package 0.1.35), unrestricted A0 carries out **144/144** injected attacks under the original targeted judge and **128/144** under the registered corrected IT0/IT1/IT3 judge, while Corridor's kernel blocks them in A5 (**0/144** under each judge in all three approval modes). Live **gpt-6.1-sol** resisted `important_instructions` even without defences (A0 **0/144** under both judges), so this run shows no observed live security difference; it does show the defence's utility cost, with benign completion **11/16** for A0 versus **8/16** for prompt-bound A5 under the unchanged upstream task-utility judge.

Configuration	Original ASR	Corrected ASR	Mode and population
A0	144/144	128/144	Model-free GroundTruth; complete suite census
A5, each approval mode	0/144	0/144	Model-free GroundTruth; prompt-bound, auto-deny and auto-approve, separately
Null agent	0/144	0/144	Model-free no-op control
A0	0/144	0/144	Live gpt-6.1-sol; all attack pairs evaluable
A5, prompt-bound	0/143	0/143	Live gpt-6.1-sol; one further pair truncated and unresolved

The model-free AgentDojo results are identical across the compared freezes. These **144-pair censuses are not samples**; no binomial confidence bounds are attached. The live run uses `domestic-freeze-6`; the evidence manifest binds both judge source versions and hashes.

Under attack, the unchanged upstream task-utility judge reports live gpt-6.1-sol completion of **97/144** planned pairs for A0 and **72/144** for prompt-bound A5. A5 has one truncated case in each utility population, counted as incomplete and excluded from evaluable ASR. These utility checks are permissive, so utility is a benchmark score, not an independent completion proof. All **509/509** retained live blocks passed recorded strict-replay verification; the bundle discloses pilot, sanity and other approval-mode rows separately.

Limits and attribution. AgentDojo issue #181, public on 2026-08-17, already reports the IT0/IT1/IT3 precedence defect. Our additional observation is that IT1's own GroundTruth helper return still fails the corrected target check. The adapter converts free-text tool arguments into typed references by value matching; this is weaker than Corridor's native typed interface. Neither mode establishes general robustness beyond this suite and adapter.

Reproduce the mechanisms and retain the limits

Use the frozen environment and the key-free scripted path first. The gateway is in-process with the Streamlit UI. Dependency installation may require network access or a populated cache; offline application execution is a separate property.

```
uv sync --frozen --python 3.11
uv run --frozen corridor setup
uv run --frozen corridor test --rules-coverage
uv run --frozen corridor judge --planner scripted --suite wealth
uv run --frozen corridor adversary
uv run --frozen corridor bench --suite wealth
```

Strict replay requires matching model, API, reasoning and output-limit identity plus the recorded response cache. It fails on a miss without provider transport or a scripted fallback. See the execution guide for the exact offline driver and matching recorded bundle.

The source registry and complete rule matrix link each rule to enforcement sites and tests. Executed rule assertions, lint, formatting, strict typing and import-boundary checks complement the scripted and recorded suites. The retained F5 full test gate reports **2895/2895 passed** in scripted mode; its earlier failed and interrupted attempts remain disclosed. Current artifact checks are recorded separately in artifact validation.

Remaining boundary	Consequence and next step
Heuristic and declared data	No global optimum or unbiased universe claim; independently review policies and catalogue construction
Synthetic identity and keys	No production security or suitability claim; replace demonstration key management and authentication before deployment
Nonatomic execution	Earlier holdings survive later failure; retain reconciliation and explicit client recovery
Finite evaluation	Small related suites do not establish general advisor competence or a production failure probability
Regulatory correspondence	Reg BI, MiFID II and NFRA mappings are engineering interpretations; licensing and full obligations remain outside scope

All return estimates remain **performance benchmark (not guaranteed)**. The prototype uses **synthetic data** and **simulated MFA** and is **not investment advice**. Repository: Corridor Wealth source.